

Lab 02 – MATH 240 – Computational Statistics

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Abstract

This experiment seek to analyze how big of an influence did different music groups have on songs, specifically how much did the bands **Manchester Orchestra**, **The Front Bottoms**, and **All Get Out** affect the song **Allentown**. By using **Essentia**, an open-source program for music analysis and synthesis, the musical components of 180 songs were isolated to be compared with the parts of **Allentown**. This method would be give more depth into the analysis and provide a better understanding of what elements are extracted from music. Through this process, the steps of data cleaning and summarization will be explored.

Keywords: Cleaning data, extracting and automating data processing

1 Introduction

Music plays such an important role in everyday lives, from creating social spaces, providing an outlet, and connecting people throughout the world. As a result, the works of different artists can be traced to find influence from older generations or their peers. The song **Allen Town** by **Front Bottoms** and **Manchester Orchestra** has influence from the two bands and **All Get Out**, so this lab seeks to explore what elements are similar and who overall has contributed the most to the creation of this song. To do this, **Essentia** was used to

You want to provide enough background to understand the context of the work. Specifically, what is the question you are addressing? If it applies, describe what information currently exists about this problem, including citations, and explain how the question you're answering complements this work.

Provide a roadmap of the structure of the paper.

2 Methods

1.

Describe the data you are working with, if applicable. Describe the specific process you will follow to answer the question at hand. This does not mean you should write something like this.

I did this and then I did that and then I did this other thing and then..., and then..., and then...

Instead, it should provide a clear and concise narrative that flows from the problem specification in the Introduction to

how you will approach answering it. This is where I would expect to see some citations for R packages you will use to conduct the statistical analysis reported in the Results section.

2.1 Methods Subsection

Much like the Introduction, subsections can be helpful for the Methods section. For example, you might describe data collection and the statistical analyses of the collected data in different subsections. Or, you may have different questions that require distinct methods.

3 Results

Tie together the Introduction – where you introduce the problem at hand – and the methods – what you propose to do to answer the question. Present your data, the results of your analyses, and how each reported aspect contributes to answering the question. This section should include table(s), statistic(s), and graphical displays. Make sure to put the results in a sensible order and that each result contributes a logical and developed solution. It should not just be a list. Avoid being repetitive.

3.1 Results Subsection

Subsections can be helpful for the Results section, too. This can be particularly helpful if you have different questions to answer.

4 Discussion

You should objectively evaluate the evidence you found in the data. Do not embellish or wish-terpet (my made-up phrase for making an interpretation you, or the researcher, wants to be true without the data *actually* supporting it). Connect your findings to the existing information you provided in the Introduction.

Finally, provide some concluding remarks that tie together the entire paper. Think of the last part of the results as abstract-like. Tell the reader what they just consumed – what's the takeaway message?

Bibliography: Note that when you add citations to your bib.bib file *and* you cite them in your document, the bibliography section will automatically populate here.

5 Appendix

If you have anything extra, you can add it here in the appendix. This can include images or tables that don't work well in the two-page setup, code snippets you might want to share, etc.